

Class 5: Modeling, Evaluation, and Adaptation

Data-driven Systems Engineering

Agenda

- Modeling as abstraction and representation choice.
- Matching model families to problem types and constraints.
- Evaluation beyond accuracy.
- Offline learning, online learning, and adaptation.
- How modeling choices affect deployment and monitoring later.

Learning goals

- Frame modeling as an engineering choice rather than a leaderboard game.
- Match model types to data, objectives, and operational constraints.
- Evaluate models using metrics that reflect the real task.
- Explain when adaptation or online learning becomes necessary.

Course Map and Running Cases



Today in the course

Focus: Model choice, evaluation, and adaptation.

Bridge: This class turns engineered features into predictive behavior and prepares the later discussion on serving, deployment, and monitoring.

Fraud case

In the fraud case, the key question is how to score rare events with metrics and thresholds that reflect business cost.

Pasteurization case

In the pasteurization case, the key question is whether forecasting or state-aware prediction helps operators anticipate abnormal behavior.

Course thread: The course thread now shifts from representation quality to decision quality and operational trade-offs.

What is modeling?

- Modeling is the process of representing patterns in data so the system can generalize to unseen cases.
- Every model encodes assumptions about structure, noise, and decision boundaries.
- Choosing a model means choosing a trade-off among accuracy, interpretability, speed, and maintainability.

Useful modeling questions

1. What decision or prediction are we supporting?
2. What kind of target do we have: class, score, or continuous value?
3. What kinds of errors are expensive?
4. How fast must training and inference be?
5. How often do we expect the data distribution to change?

Linear models	Fast, simple, and often strong baselines.
Tree-based models	Good for heterogeneous tabular data and nonlinearity.
Probabilistic models	Useful when uncertainty and conditional structure matter.
Neural networks	Powerful for large unstructured or highly nonlinear problems.
Online models	Update incrementally as new observations arrive.

More realistic examples

- Fraud detection: random forest or gradient boosting with threshold tuning.
- Streaming weather prediction: online linear regression with River.
- Pasteurization forecasting: incremental regression plus drift alerts.
- Industrial anomaly detection: useful only if alert cost and false positives are understood.

Evaluation must match the task

- Imbalanced classification needs precision, recall, PR-AUC, and threshold analysis.
- Regression needs error magnitude metrics such as MAE or RMSE.
- Calibrated probabilities matter when downstream action depends on confidence.
- Operational cost matters as much as numerical score.

Repository example: fraud model training pipeline

```
X = df.drop("is_fraud", axis=1)
y = df["is_fraud"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

- 'streamlit/modeling.py' gives a concrete tabular classification flow.
- Stratification matters because fraud is rare and naive splits can distort the class balance.

Fraud dataset reality: extreme class imbalance



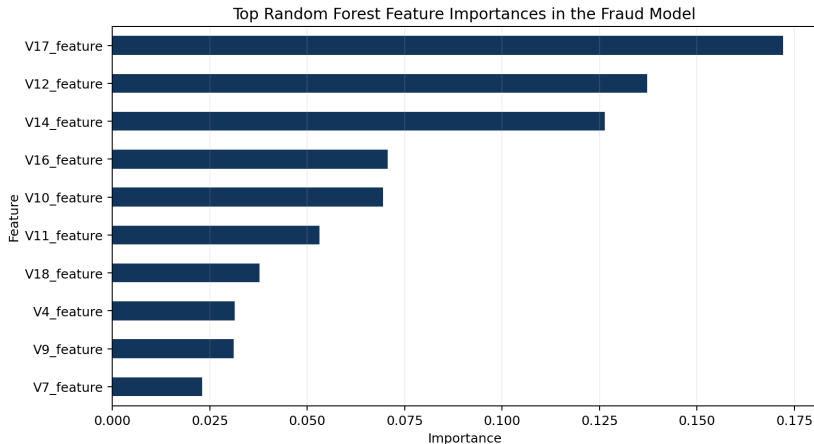
From 'Credit_Card_Fraud_Detection_Solution.ipynb': the first modeling challenge is not algorithm choice but the rarity of the positive class.

Repository example: fit, predict, and inspect errors

```
cls = RandomForestClassifier(  
    n_estimators=200, random_state=42, n_jobs=-1  
)  
cls.fit(X_train_scaled, y_train)  
  
y_pred = cls.predict(X_test_scaled)  
print(confusion_matrix(y_test, y_pred))  
print(classification_report(y_test, y_pred, digits=4))
```

- This is a better discussion anchor than accuracy alone because the confusion matrix exposes missed fraud cases directly.
- The classification report opens the door to threshold and cost-sensitive evaluation.

Model interpretation: top feature importances



Generated from the saved 'streamlit/model.pkl': tree-based models can highlight influential variables, but importance is still not the same as causality.

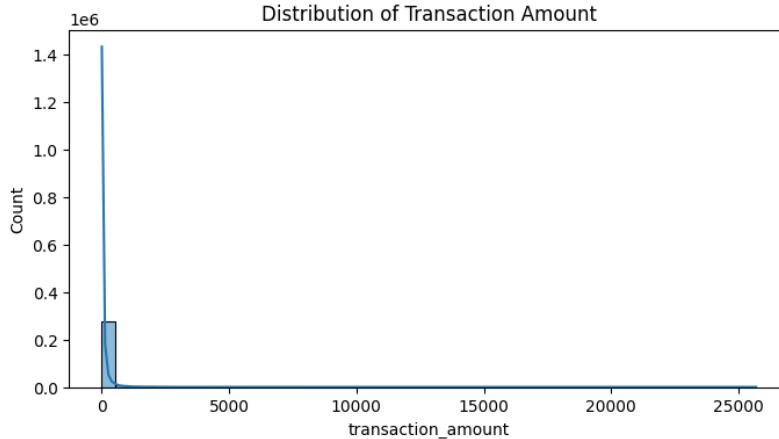
Why accuracy can fail badly

Fraud example

If only 0.2% of transactions are fraudulent, a model that predicts “legitimate” every time may look highly accurate while being operationally useless.

- This is why threshold, recall, and precision are part of the engineering conversation.

Distribution shift starts with the raw variables



The transaction amount distribution from the fraud notebook is highly skewed, which is a useful reminder that modeling choices interact with preprocessing and scale.

Offline, online, and adaptive learning

Offline

- Train on a fixed dataset
- Easier benchmarking
- Periodic retraining

Online

- Update one sample at a time
- Works on streams
- Responds faster to change

Repo activity

'notebooks/Class8_Model_Incremental_learning.ipynb' and 'streamlit/pages/App_Stream.py' show how incremental learning changes both evaluation and monitoring.

Why adaptation becomes necessary

- User behavior changes.
- Sensors drift or get recalibrated.
- Business policies or regulations shift.
- New classes of errors appear after deployment.

Key point

Model adaptation is not a sign that the first model failed. It is a recognition that the environment evolves.

Streaming example: incremental learning with River

```
model = (  
    preprocessing.StandardScaler()  
    | linear_model.LinearRegression(optimizer=optim.SGD(0.01))  
)  
mae = metrics.MAE()  
  
y_pred = model.predict_one({"wind": wind})  
model.learn_one({"wind": wind}, temp)  
mae.update(temp, y_pred)
```

- This pattern appears in 'streamlit/pages/App_Stream.py'.
- It makes adaptation tangible: the model learns one observation at a time instead of waiting for a full retraining cycle.

Papers and materials

[scikit-learn documentation](#); [scikit-learn model evaluation guide](#); [River documentation](#); [River metrics API](#).

From This Class to the Next

What stays with us

- Model selection is always tied to the prediction task, the error costs, and the available data.
- Metrics are decision tools, not decorative scoreboards.
- Offline training and online adaptation lead to different operational behaviors.

Project use

This is the point where teams can defend a baseline, compare alternatives, and explain why a chosen model fits the constraints of the fraud or pasteurization setting.

Next connection

Class 6 shows how trained models become usable through APIs, dashboards, and human-facing interfaces.

Course thread: A model becomes valuable only when its outputs can be consumed, inspected, and acted on in a real workflow.

Papers and materials

Bishop, Pattern Recognition and Machine Learning; Mitchell, Machine Learning; Gama et al., A survey on concept drift adaptation; Domingos (2012), A Few Useful Things to Know About Machine Learning.